Setting Up a Research Project

ECON 895 Spatial Techniques in Empirical Economics Week 6

Start from the template. The template is public. Copy it into a new folder for your paper and give it its own history.

```
git clone https://github.com/noeldjohnson/research-workflow-template_shared.git my-paper
cd my-paper
rm -rf .git && git init
```

Before the first session with a model. Read `.claude/settings.json` and the scripts in `.claude/hooks/`. Claude Code runs those scripts on your machine with your permissions, and the settings file lists commands the model may run without asking. Do this for any repository you clone, including this template. Then make a first commit, so every later change shows up in `git diff`.

What each folder is for. Work flows one way, from raw data to the paper, and everything after `Data/raw/` can be rebuilt by code.

`Data/raw/`	Data as downloaded. Never edited, by you or by a model.
`Data/processed/`	Cleaned data, written only by scripts.
`scripts/`	R scripts, numbered and run in order, plus one master script that runs them all.
`output/`	Intermediate results saved as `.rds`.
`Tables/`, `Figures/`	Written by scripts and read by the paper. Never edited by hand.
`Paper/`	The manuscript. `main.tex` pulls in `sections/`, tables, and figures.
`notes/`	Reading notes and research notes.
`explorations/`	Exploratory work, held to a lower standard than the paper.
`quality_reports/`	Plans, dated session logs, and review reports.

What the model reads.

- `CLAUDE.md`, at the start of every session. Fill in the research summary and keep the project-state table current.
- `.claude/rules/`, conventions that always apply. A `paths` line limits a rule to some files, such as R scripts.
- `.claude/agents/`, specialist reviewers with their own instructions. A tools line such as `Read, Grep, Glob` keeps one read-only.
- `.claude/skills/`, procedures you run by name, such as `/data-analysis` or `/review-paper`.
- `.claude/hooks/`, scripts Claude Code runs at fixed moments whatever the model decides. The model can fail to follow a rule. It cannot skip a hook. One hook can make `Data/raw/` read-only.

Habits that hold with or without a model.

- Raw data is read-only. A `README` records each dataset's source, version, and download date.
- Every number in the paper comes from a script, and one command rebuilds them all.
- Paths are relative to the project folder, and random draws are seeded once.
- The project lives in git from the first day. Commit before every session with a model and read `git diff` after it.
- A dated log records what you did and what you decided.

A project organized this way already holds most of a replication package, meaning the data with its provenance, code that runs in order, and the outputs the paper reports.

Optional before Week 7, ungraded. Clone the template for your paper and fill in the research summary in `CLAUDE.md`, meaning the question, the hypotheses, the data strategy, and the setting. Bring it to office hours if you want a second reader.